Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

Evaluating Frontier LLM Translation Capability for Lakota

Lance Robertson · University of California, San Diego

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No frontier LLM reliably translates Lakota.

Even measuring how badly they fail is hard.

Lakȟótiyapi

the language under test

Why test this?

Lakȟótiyapi (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today's LLMs list translation support across dozens of languages, but providers don't say what they can or can't do per language. A reasonable user can't tell Lakota apart from any of the others on the list — *if it does all these, why not this one?* Direct evaluation is the only way to find out.

Is it just memorized?

The evaluation set comes from a published dictionary, so contamination is the obvious worry. The Lakota source appears verbatim online for ∼10% of pairs, but the Lakota and its English reference together for only 0.5%. Scores spread widely (6–61% equivalence), not clustered at ceiling — the signature of partial capability, not recall.

Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions**.
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF++; diacritic-normalized chrF++; an LLM *semantic* judge on the English side.

LLM-as-judge for L → E

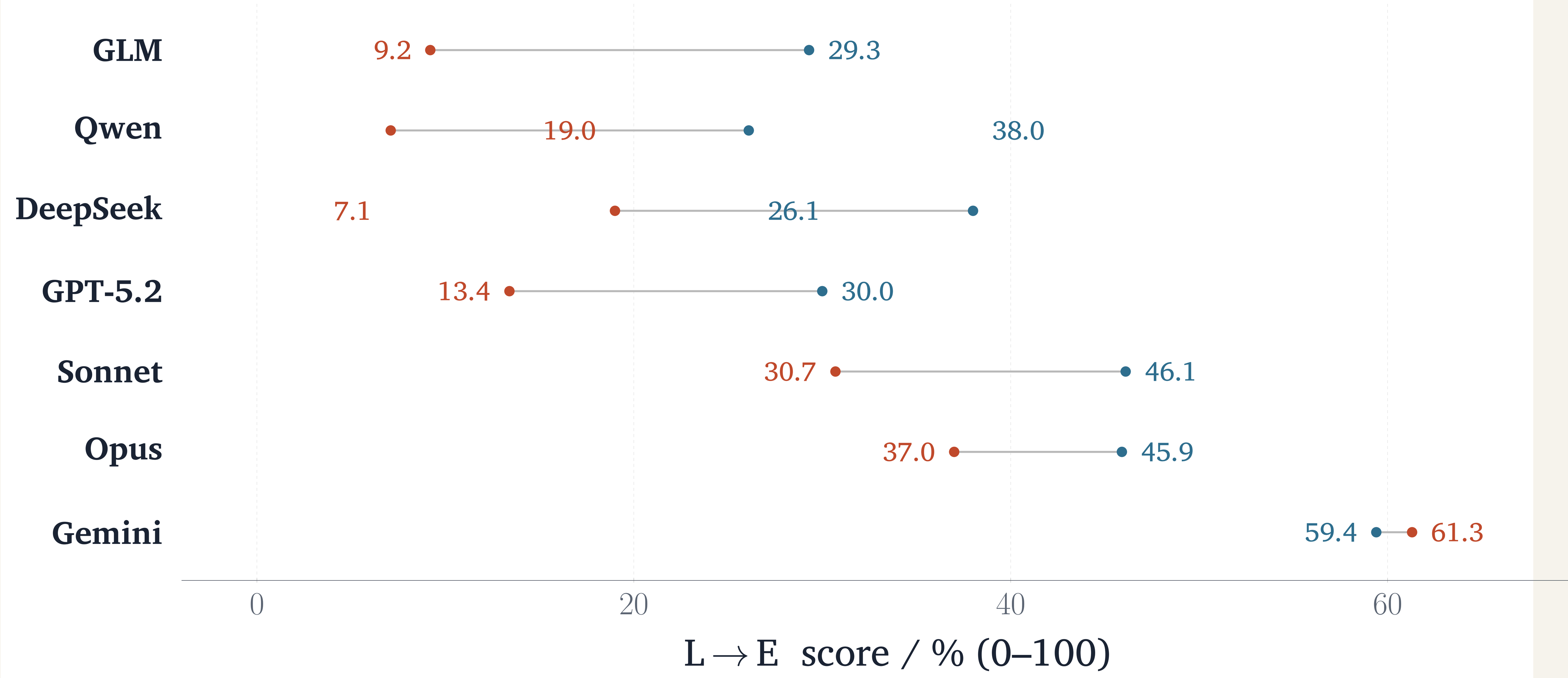
chrF++ measures character and word overlap, but it can reward plausible nonsense. We wanted an automated measure that could take meaning into account. We use an LLM-as-judge: for the L → E direction the output is English, so a model can assess whether each translation carries the reference's meaning. To keep the judge from favoring its own family, every pair is scored by two LLMs from different families, Gemini 3 Flash and an independent Llama-3.3-70B. They agree substantially (Cohen's $\kappa = 0.75$, 0.89 quadratic-weighted), so the judgments are cross-validated rather than one model's unchecked read.

Standard Lakota Orthography

ȟ ą į ų á ŋ ’

SLO encodes phonemic distinctions with diacritics; stripping them adds **+3.3 to +6.5 chrF++** (E → L) — models get base characters right, marks wrong.

Surface metrics overstate capability



Each row is one model, ranked by gap size (largest at top). Connector length is how far the surface score chrF++ (steel) overstates % meaning-equivalent (terracotta). Only Gemini's two scores nearly agree.

Translation quality (chrF++)

Model	L → E	E → L
Proprietary		
Gemini 3.1 Pro	**59.4**	**42.6**
Claude Opus 4.6	45.9	34.3
Claude Sonnet 4.6	46.1	27.0
GPT-5.2	30.0	23.6
Open-weight		
DeepSeek-V4-Pro	38.0	29.2
GLM-5.1	29.3	14.6
Qwen3.6-Plus	26.1	18.4

Best condition per model. Every model scores higher *out of* Lakota than *into* it.

Semantic equivalence (L → E)

Model	Equiv.	Partial	Not
Gemini 3.1 Pro	**61.3**	23.1	15.6
Claude Opus 4.6	37.0	21.0	42.0
Claude Sonnet 4.6	30.7	23.8	45.5
DeepSeek-V4-Pro	19.0	14.0	67.0
GPT-5.2	13.4	9.8	76.8
GLM-5.1	9.2	6.5	84.2
Qwen3.6-Plus	7.1	7.6	85.4

% of judged pairs, reasoning condition.

How models fail

Reference	Model	Output	chrF++
We live in a small trailer house with no electricity.	DeepSeek	"small *sacred* tipi"	18.6
Make coffee (*Wakȟálya yo*).	DeepSeek	"Sanctify it!"	2.9
Did my daughter call?	GLM	"Did my younger brother come?"	27.6
Do you have a car?	GPT-5.2	"Do you have a horse?"	66.4

Idioms and coined terms are translated *element by element*: *Wakȟálya yo* ("make coffee," literally "make it boil") becomes "sanctify it." A one-word near-miss — "car" rendered "horse" by a *frontier* model — still scores 66.4 chrF++. Surface overlap and meaning preservation diverge.

Phrasebook concentration (L → E)

Model	Sc. 4	Sc. 5	Sc. 6	Δ
Gemini 3.1 Pro	52.7	58.2	**81.9**	+29.2
Claude Opus 4.6	41.6	41.0	65.6	+24.0
Claude Sonnet 4.6	40.3	45.1	64.1	+23.8
DeepSeek-V4-Pro	33.0	33.9	59.6	+26.6
GPT-5.2	27.9	23.7	45.7	+17.8
Qwen3.6-Plus	23.3	22.1	41.7	+18.4
GLM-5.1	27.3	26.0	41.0	+13.7

chrF++ stratified by NLD conversational score: 4 = more formal/complex, 6 = most conversational. Every model lifts sharply at 6 — performance concentrates on high-frequency phrasebook-like items rather than reflecting general capability.

Takeaways

- No frontier LLM reliably translates Lakota, either direction (2026).
- chrF++ overstates low-resource capability — pair it with a meaning check.
- Open-weight doesn't close the gap.

Extended reasoning changes refusal behavior more than quality (open-weight E → L: **0 → 36** refusals when enabled).

Origin

This evaluation grew out of a long-running interest in the possibility of crowdsourcing a Lakota training corpus. It started as a User Interface design problem: what could community-driven tools look like for gathering training data for indigenous-led AI? In a seminar, faculty gave us the provocation: that today's LLMs are good enough you could just *"vibe-code"* a Lakota translator over a weekend. I wanted to investigate if this was feasible. My background in software development and LLM evaluations drove me to create this Lakota benchmark.

Data & code

github.com/robotson/lakota-translation-benchmark